

This work (LQ20839A) systematically investigates the singularity structure of higher-order exceptional points (EP_n) in non-Hermitian systems. It proposes and employs the notion of “symmetry-allowed Hessenberg forms” as an analytical framework to explain the algebraic order of eigenvalue splitting under perturbations for different symmetry classes (P, C, PT). Starting from a Jordan block and using Puiseux series expansions, the authors show that the leading-order splitting of eigenvalues near an EP_n scales as $\epsilon^{1/k}$. The manuscript is well structured, mathematically rigorous, and supported by concrete models (e.g., Hopf semimetal, pseudospin-1 model) illustrating EC_3 and ES_3 structures. Overall, the work provides significant theoretical depth and systematic insight, making it suitable for publication in Physical Review A.

However, several points regarding the underlying mechanism — in particular, why the system still exhibits singular behavior after eigenvalue splitting — require further clarification. Detailed comments are provided below.

1. Why is the system still considered “exceptional” after eigenvalue splitting?

The paper emphasizes that after applying a small perturbation ϵ near an EP_n , the eigenvalues split as $\epsilon^{1/k}$. This splitting does not automatically imply that the system remains an EP. An EP is defined by both eigenvalue degeneracy and a Jordan block (defective eigenvectors). For $\epsilon=0$, the system generally becomes non-degenerate and therefore is no longer an EP in the strict sense. What exactly does the author mean by “still singular”? Does it refer to the fact that $\epsilon=0$ remains a branch point in the parameter space? Or does it mean that for some fine-tuned directions the system retains partial degeneracy or Jordan structure even for $\epsilon=0$? It is recommended to explicitly distinguish in Sec. II or Sec. V: The EP at $\epsilon=0$ is singular. For $\epsilon=0$, the system is generally no longer an EP, but the eigenvalues as functions of ϵ exhibit a branch-point singularity at $\epsilon=0$.

2. The paper shows that by choosing specific parameters (e.g., a_4 in the P-symmetric system), the $\epsilon^{1/2}$ term vanishes and the leading behavior becomes $\sim \epsilon$. In such cases: Does the original EP_3 persist? Does it degrade to an EP_2 or a Dirac-like crossing? Does this direction-dependent behavior imply a higher-codimension singular structure in parameter space? This mechanism is crucial for understanding EP stability and for sensor design. A brief geometric picture or a simple example would be helpful.

3. The paper shows that P and C symmetries only allow upper-2 Hessenberg, while PT allows upper-3 Hessenberg. This is a central result, but the argument relies on explicit matrix forms. A more abstract group- or representation-theoretic explanation would be beneficial. How do commutation relations with the symmetry operator enforce the specific upper-triangular structure in the Jordan basis? Can this be generalized to higher dimensions ($s>4$)?

4. In Sec. V, direction-dependent sensitivity is mentioned for EP-based sensors. However, if the system is no longer at an EP after perturbation, does the enhanced

sensitivity remain? Sensors typically operate near an EP, not exactly at it. The branch-point singularity ($\epsilon^{1/k}$) is the source of sensitivity, even when the system is non-degenerate for $\epsilon \neq 0$.

5. The manuscript repeatedly mentions “topological phases”, “topologically robust phenomena”, and “symmetry-protected nodal phases”, but it does not present any analysis or evidence of a topological phase transition. Specifically: No topological invariant (Chern number, winding number, biorthogonal polarization, etc.) is computed or compared across parameter regions. The existence of an EP does not by itself constitute a topological phase transition; a transition requires gap closing/reopening accompanied by a change in a topological invariant. The P- and C-symmetric models always host a flat zero-energy band, and the PT-symmetric models are not analyzed in terms of invariant changes. Therefore, the current manuscript does not address topological phase transitions. The author should clarify it.

6. In Page 4, “ $f_1 =$ and $f_2 = 0$ ”, what is f_1 ? In Sec V, “cna” should be “can”.